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Health and Happiness

Investigating the Impact of Significant Health Conditions on Well-Being

Abstract

Despite significant research on happiness, its relation to health conditions contains gaps. Yet understanding how happiness is related to health conditions is critical to improving healthcare and improving quality of life for those with medical conditions. This study aimed to examine whether a significant health condition, defined as an impairment that substantially impedes an individuals' ability to perform daily tasks, was correlated with happiness levels, which was measured on a scale from 1 to 7. Data was collected from 755 individuals through a google form and analyzed using a chi-square test for independence. The test resulted in a p-value of 0.43959 and showed no correlation between happiness and health. This invites a re-examination of factors for happiness and suggests resilience may help those with health conditions adapt so they have a similar scale of happiness to those without health conditions. Future research should focus on resilience, specifically hedonic adaptation, and its role in those with health conditions.

Keywords: Health conditions, Happiness, Health impairment, Resilience

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Background

From Hippocrates in Ancient Greece to modern studies like the Harvard Study of Adult Development, the relationship between health and mental well-being has been continuously explored and tested. Research has proven that health and happiness are positively correlated (ie. healthier people tend to be happier) (American Heart Association 2022). However, these studies only consider physical health (ie. exercise, diet, and other controllable factors). The studies that do consider disabilities tend to focus on neurodegenerative diseases, which make up 15% of the global population (Van Schependom et al. 2019), rather than general health conditions such as blindness, mental health conditions, and mobility disabilities.

Significantly limiting health conditions is defined by the NIH as a physical or mental impairment that substantially impedes an individuals' ability to perform daily tasks (National Academy of Sciences 1999). Approximately 16% of the global population is affected by these health conditions. Because such a large proportion of the world is affected, it is important to understand the relationship between health and happiness to create a better healthcare system and a society better equipped to support these individuals.

Introduction

16% of the global population, or approximately 1.3 billion people, experience a health condition that significantly limits their quality of life (WHO). In recent decades, the global population has aged significantly, resulting in a greater proportion of the world with debilitating health conditions. At the same time, global happiness levels have been decreasing, from 77% in 2011 to 70% in 2020 (Stinson 2024). Understanding the relationship between health and emotional well-being is essential to improve quality of life for those with medical conditions.

Beyond physical limitations, severe health conditions often lead to feelings of social exclusion, reduced independence, and psychological anxiety. The relationship between health and happiness can help inform healthcare and public health policies to more effectively address the holistic needs of individuals. This study aims to add to knowledge about the presence of a relationship between health and happiness to improve societal support for individuals affected by health conditions. We hypothesize that a significantly limiting health condition will be associated with lower levels of happiness. Such conditions can impose substantial restrictions on daily life and constantly reinforce beliefs of not fitting in with the rest of society, which takes a substantial mental toll that may result in lower levels of happiness.

Methods

Data for this study were collected through a structured survey consisting of multiple-choice questions. The survey was designed to assess individuals' happiness levels and whether they had significantly limiting health conditions. Participants from the general population aged 13 and above were invited to complete the survey, which was distributed online to ensure accessibility and a diverse sample.

A total of 755 responses were collected. The survey included demographic questions, as well as targeted questions aimed at identifying patterns, preferences, and opinions relevant to the research question. To analyze the data, responses were compiled and examined using quantitative, statistical methods with the goal of identifying correlations between variables of interest.

In addition to survey data, external sources and relevant literature were used to provide context and support for the findings. These sources helped to triangulate the data and strengthen the interpretation of observed trends.

Results

We are looking at whether there is a correlation between happiness levels and whether there was a medical condition with a significant impact on the individuals' lives.

27% of respondents said they had a medical condition that had a major impact in their life, while 73% said they did not. Happiness level was measured on a scale of 1 to 7, with 1 being unhappy on average and 7 being very happy on average. The median happiness level was 5, with an average of 4.539. The standard deviation was 1.485. The distribution for happiness level was approximately symmetrical and unimodal, with the most common response being 4 and had no outliers.



Figure 1: Distribution of happiness level

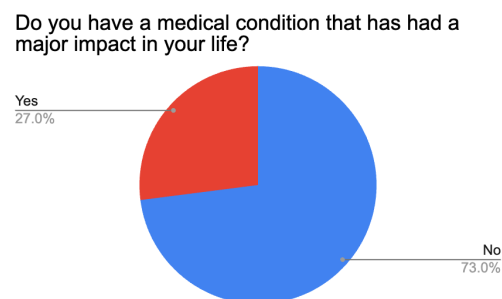


Figure 2: Distribution of health conditions

We divided the data into two categories based on whether they had a significant health condition, and compared the distribution of happiness levels between the two groups. The sample of those with a significant health condition had 204 respondents, a mean of 4.5637 and a

standard error of 1.7309. The sample of those without health conditions had 551 respondents, a mean of 4.5535 and a standard error of 1.4833.

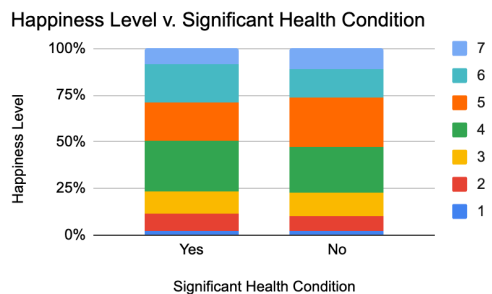


Figure 3: Segmented bar graph of distribution of happiness level based on presence of significant health condition

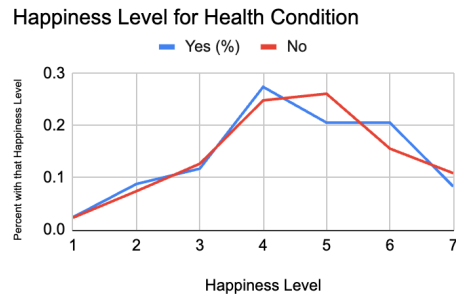


Figure 4: Line chart of distribution of happiness level based on presence of health condition

To test for a significant difference in distribution of happiness levels between those with and without health conditions, we performed a χ^2 -test for homogeneity at the 0.05 significance level.

		Has Significant Health Condition	
		Yes	No
Happiness Level	1	5	13
	2	18	41
	3	24	70
	4	56	137
	5	42	144
	6	42	86
	7	17	60

Figure 5: Table of happiness level distribution based on presence of significant health condition

Hypothesis:

H_0 : There is no difference in distribution of happiness levels between those with and without significant health conditions.

H_a : There is a difference in distribution of happiness levels between those with and without significant health conditions.

The resulting χ^2 value was 5.8553 and the p-value was 0.43959. Since $0.43959 > 0.05$, we fail to reject the null. There was not convincing evidence that there was a statistically significant difference in the distribution of happiness level between those with health conditions and those without health conditions.

Discussion

The data shows that there is no correlation between the presence of a significant health condition and happiness level. Using a χ^2 -test for independence, we found a p-value of 0.43959: this means that the differences in happiness level distribution observed in the sample is 43.959% likely to occur purely because of chance. At a significance level of 0.05, this does not represent a significant difference; we cannot conclude that there is a significant difference between distribution of happiness levels between those with and without health conditions. When looking at Figure 3 and Figure 4, we are able to visually see that there is very little difference in the distributions of happiness levels between those with and without significant health conditions.

Previous studies have shown that better physical health, including regular exercise and the absence of chronic illness, is often linked to higher happiness and life satisfaction (Veenhoven 2008). Considering the results of the study, the lack of a significant relationship in our study may point to the influence of other factors like psychological resilience or social support in maintaining well-being.

A possible reason for the lack of relationship may occur because of hedonic adaptation: individuals tend to return to a baseline level of happiness despite changes in life circumstances (Sylvester et al. 2018). Psychological resilience allows many individuals to maintain or recover their well-being despite ongoing health challenges. Multiple studies have shown resilience positively influences life satisfaction and happiness for people with chronic illnesses (Tecson et al. 2019). With time and resilience, people with significant health conditions may adjust their expectations and coping strategies, allowing them to experience levels of happiness comparable to those without such conditions.

The research does include several limitations. Since the study was observational, only correlation can be assumed, not cause and effect. In addition, the sample was a convenience sample since we only surveyed friends and family members of CIR4 members, which prevents us from generalizing this conclusion to the general, global population. The results were self-reported so there may also be self-reporting bias, where respondents reported a happiness level or health condition significance that may not be objectively accurate.

The finding of no correlation between significant health conditions and happiness suggests that well-being may not be as directly affected by health status as commonly assumed. This invites a re-examination of the factors that truly drive happiness beyond health. Future research might benefit from more random and representative sampling to generalize the conclusion to a greater population. Ethical experiments in the future, if possible, would allow us to determine whether there is a causal relationship. If happiness remains similar regardless of health condition, interventions to improve well-being may focus more on other factors such as psychological resilience, social support, and life skills rather than just physical health management.

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